

MOD300 Anvendt Python programmering og modellering

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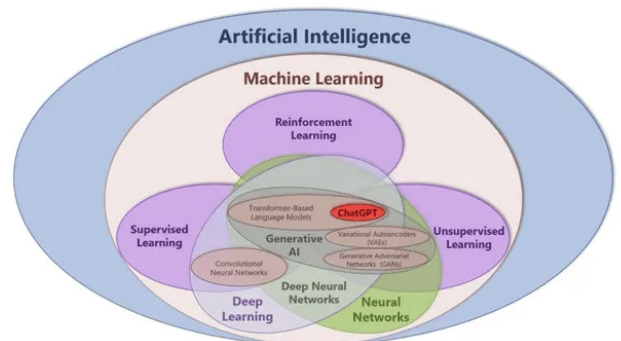
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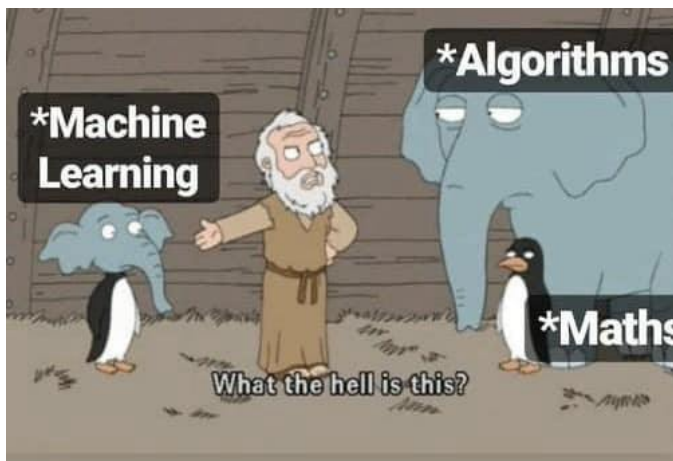
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Statistics, Machine learning or Artificial intelligence?

What is the main difference between the three fields?



How Machine Learning Started?



Statistics

Let's start from the definition

- Statistics (origin "description of a state/country") is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data.
- It is conventional to begin with a statistical population or a statistical model to be studied. Populations can be diverse groups of people or objects such as "all people living in a country" or "every atom composing a crystal".
- Statistics deals with every aspect of data, including the planning of data collection in terms of the design of surveys and experiments.[Wikipedia]

Machine learning

Definitions:

- Machine learning is a branch of artificial intelligence (AI) and computer science which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy. [IBM]
- Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data, and thus perform tasks without explicit instructions. [WIKI]
- Machine learning is a subfield of artificial intelligence that uses algorithms trained on data sets to create models that enable machines to perform tasks that would otherwise only be possible for humans, such as categorizing images, analyzing data, or predicting price fluctuations. [Coursera]

Machine learning

One technical definition

Machine learning is a set of computer based statistical approaches that aim to minimise the loss function to maximise inference accuracy. [Enrico, 5.2.2024]

The loss function is the actual engine in machine learning.

Loss function

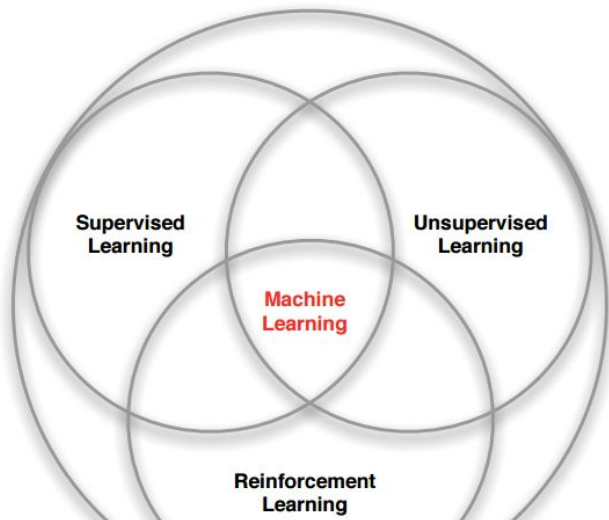
It quantifies the difference between the predicted outputs of a machine learning algorithm and the actual target values.

Back to ML

Artificial intelligences

And more definitions:

- Artificial intelligence is the intelligence of machines or software, as opposed to the intelligence of humans or other animals. It is a field of study in computer science that develops and studies intelligent machines. [WIKI]
- Artificial intelligence (AI) is the theory and development of computer systems capable of performing tasks that historically required human intelligence, such as recognizing speech, making decisions, and identifying patterns [Coursera]
- It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence, but AI does not have to confine itself to methods that are biologically observable. [IBM]



Model evaluation

We can measure the accuracy of a prediction only when we have a 'truth' to compare with.

Model outcome

The accuracy of a prediction is limited by the available data set (not an universal quantity).

Given a dataset, we split the data in a **train** and a test set.

One has to be extremely careful to not introduce a **bias** in each of them when splitting. (or to introduce a bias properly, when for example, working with time series).

Train

A train dataset is a subset of the original data. Generally it is circa a 70 % (common practice, not a rule) of the *INSTANCES*.

For each selected instance, all the features are kept, as their labels.

Random measurements

Some models are able to also consider the sequence of instances; the selection of the **train** dataset has to be randomised.

The opposite consideration shall be done for time series, where each data point is correlated to the previous.

Validation

A validation dataset is a subset of the original data. Generally it is circa a 15 % (common practice, not a rule) of the *INSTANCES*.

It is aimed to provide a **unbiased estimate of the model prediction error**. This information is then used to *tune* the model set up (indirectly). I.e. it is not independent from the test phase.

For each selected instance, all the features are kept, as their labels.

Random measurements

Some models are able to also consider the sequence of instances; the selection of the **validation** dataset has to be randomised.

The opposite consideration shall be done for time series, where each data point is correlated to the previous.

Test

A test dataset is a subset of the original data. Generally it is circa a 15 % (common practice, not a rule) of the *INSTANCES*.

For each selected instance, all the features are kept, as their labels.

Random measurements

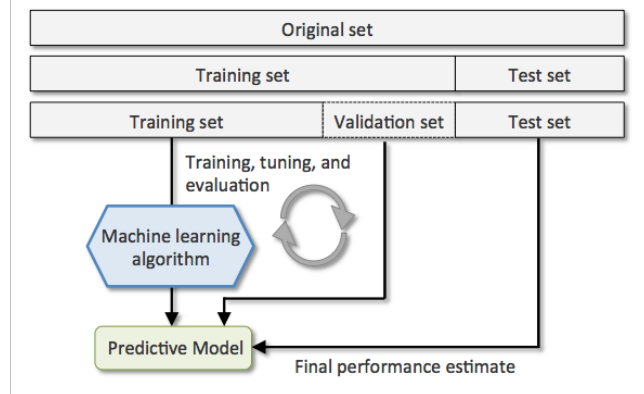
Some models are able to also consider the sequence of instances, the selection of the **test** dataset has to be randomised.

Model evaluation

Train-Validation-Test

- 1 Data is split into train, validation and test dataset
- 2 A model is trained on the train dataset
- 3 The model (hyper)parameters are improved on the validation dataset
- 4 The model performances (e.g. accuracy) are tested on the test dataset

Model evaluation

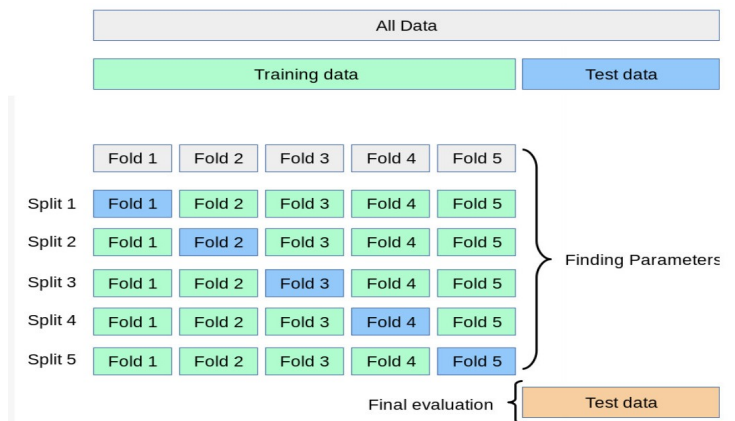


Cross validation

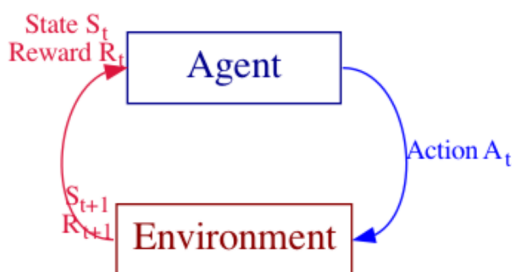
Cross validation-Test

- 1 Data is split into train and test.
- 2 Train dataset is then split into k-folds (different subsets, usually 5 to 10)
- 3 A model is trained on all k subset but one, sequentially
- 4 The model is tested on the remaining k data subset and its output averaged
- 5 The model performances (e.g. accuracy) are tested on the test dataset

Cross validation



Reinforcement learning



Main definition

Loop of

State, action, reward

- 1 You evaluate the **state** where you are
- 2 you pick an **action**
- 3 you get a **reward**



Environment, State, Policy, Agent, Reward

Environment
Physical world in which the agent operates

State
Current situation of the agent

Agent
Decision Maker that take action according to a policy

Policy
Method to map agent's state to actions
 $\pi(a|s) = P(A_t = a|S_t = s)$

Reward
Feedback from the environment

Episode
As Series of Atomic Experiences

Math backbone

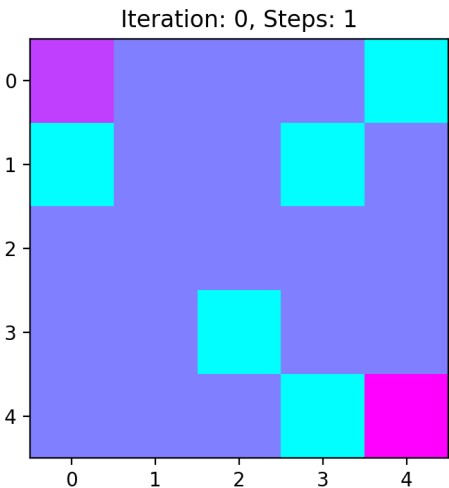
Q-learning

$$Q^{new}(S_t, A_t) = (1 - \alpha)Q(S_t, A_t) + \alpha(R_{t+1} + \gamma \max_{\alpha}(S_{t+1, \alpha}))$$

where:

- α is the learning rate
- γ is the discount factor
- $\max_{\alpha}(S_{t+1, \alpha})$ is the estimate of the optimal future value
- R is the reward
- A is the action
- S is the state

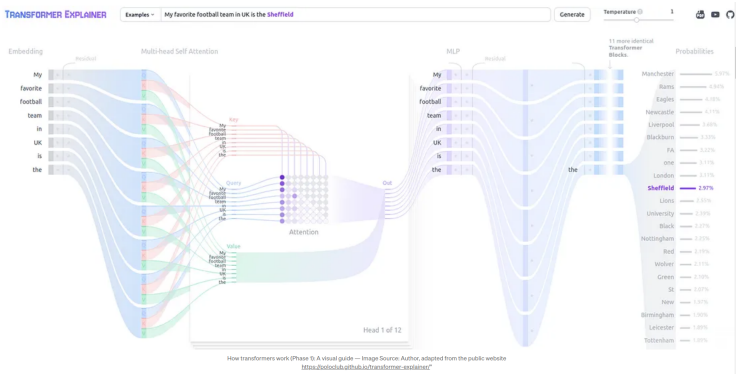
Example of RL



RL in LLMs

Pre-training

First goal: the next word prediction



RL in LLMs

Supervised Fine-tuning (SFT)

Language is inherently flexible - there are many valid ways to respond to any prompts

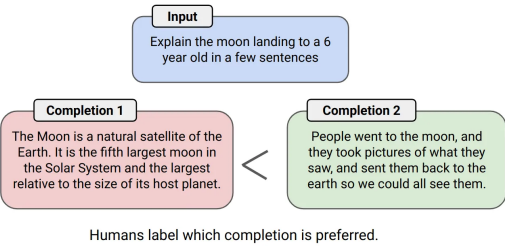
Many responses have been provided by humans. Multiple trainers in multiple countries.

Prompt	Response
Serendipity means the occurrence and development of events by chance in a happy or beneficial way. Use the word in a sentence.	Running into Margaret and being introduced to Tom was a fortunate stroke of serendipity.
ELIS: What's the cause of the "anxiety lump" in our chest during stressful or disheartening experiences?	The anxiety lump in your throat is caused by muscular tension keeping your glottis dilated to maximize airflow. The clenched chest or heartache feeling is caused by the vagus nerve which tells the organs to pump blood faster, stop digesting, and produce adrenaline and cortisol.
Create a shopping list from this recipe: Trim the ends off zucchini. Cut zucchini in half lengthwise; scoop out pulp, leaving 1/2-in. shells. Finely chop pulp. In a skillet, cook beef, zucchini pulp, onion, mushrooms and peppers over medium heat until meat is no longer pink; drain. Remove from the heat. Add 1/2 cup cheese, ketchup, salt and pepper; mix well. Spoon into the zucchini shells. Place in a greased 13x9-in. baking dish. Sprinkle with remaining cheese.	Zucchini, beef, onion, mushroom, peppers, cheese, ketchup, salt, pepper

RL in LLMs

Human feedback

Comparative feedback between answers.



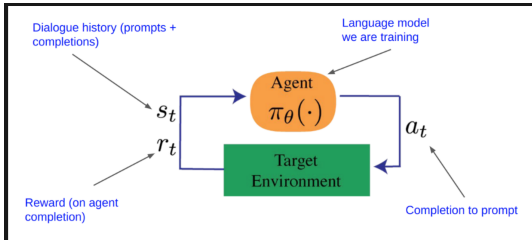
Humans label which completion is preferred.

State, Evaluate, Reward loop in LLMs

The agent is the LLM: It makes a few prompt

Human say which one is the best

LLMs generates the text such that maximize the feedback



Classification

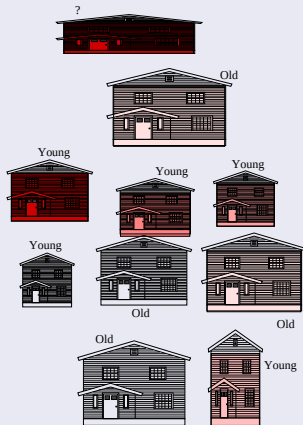
- UnSupervised classification is about “non-guided” **discrimination**
- Supervised classification is about “guided” **discrimination**
- Each object/sample has **external information** (or class membership)

Aim:

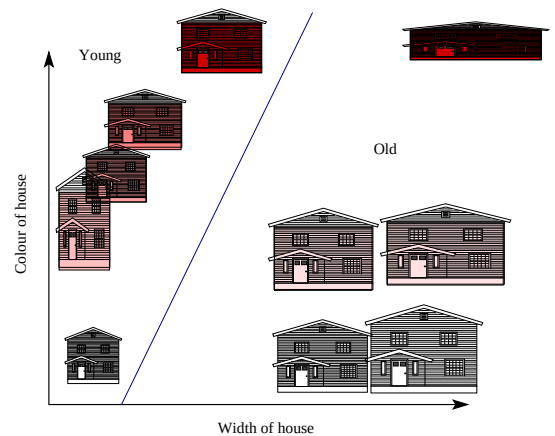
- To find a **rule** for the membership of new objects
- Identify **what attributes** are important

Classification

Do we see any rule?

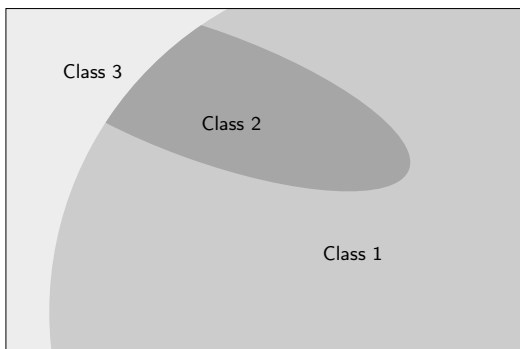


Partitioning the attribute space



Decision boundaries

Many classification methods can be viewed as a **partitioning** of the feature space such that each partition of space represents a **class**. Partitioning performed by **decision boundaries** (can be linear or curved)



Discriminant functions

Probability functions behave like a **classifier** which assign a class to a new sample object: $\mathbf{x}$ to class c_i if:

$$g_i(\mathbf{x}) > g_j(\mathbf{x}), \quad \forall i \neq j$$

The **decision surfaces** resulting from using discriminant functions are defined as:

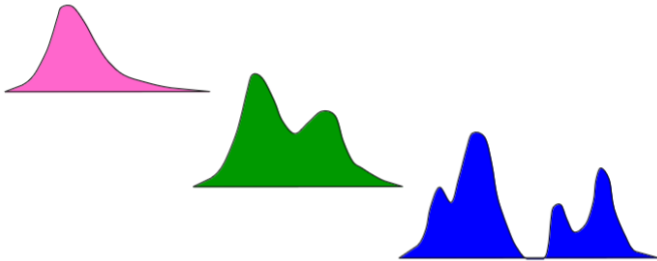
$$g_{ij}(\mathbf{x}) \equiv g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0, \quad i \neq j.$$

Probability distribution

A metric is needed

Amount of uncertainty

Number of Modes: unimodal, bimodal, polymodal



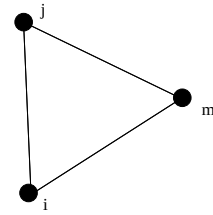
Metrics

Triangle inequality

Considering a vectors in an N-dimensional space, to be a **distance** it must satisfy the **triangle inequality**:

$$d_{ij} + d_{im} \geq d_{jm}$$

If also $d_{ij} = 0$, if $d_{ij} - d_{ji} = 0$, then we call it a *metric*.



Metris example

Common metrics:

- Counting
- Euclidean

$$d_{ij}^{(E)} = \left[\sum_{k=1}^N (x_{ik} - x_{jk})^2 \right]^{\frac{1}{2}}$$

Categorical variables do not necessarily satisfy these relations.